

Differentiating inverse trig functions

Key Points

To differentiate inverse trig functions rearrange the question, differentiate, find the reciprocal and substitute back

e.g. for $y = \arcsin x$ so $x = \sin y$ then find $\frac{dx}{dy} = \cos y$ and so $\frac{dy}{dx} = \frac{1}{\cos y}$ and to sub $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$

$$\text{So } \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$

Consideration of the domain can be important to know which sign

Basic Skills

Q1. Write down the derivative of $3 \sin 2x$

Q2. Given the exact value of $\sin \theta = \frac{2}{3}$ find the exact value of $\cos \theta$

Q3. Given the exact value of $\sec \theta = 5$ find the exact value of $\tan \theta$

Q4. For $y = \cos^{-1} x$

a) By writing x as a function of y find $\frac{dx}{dy}$

b) Hence show that $\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$

c) Hence find the gradient of $y = \cos^{-1} x$ at $x = 0$

Competency Skills

Q5. Find $\frac{d}{dx}(\tan^{-1} x)$

Q6. Find $\frac{d}{dx}(\sec^{-1} x)$

Q7. Show that $\frac{d}{dx}\left(\arcsin\left(\frac{x}{a}\right)\right) = \frac{1}{\sqrt{a^2-x^2}}$ by

a) Using $x = a \sin y$

b) Using the chain rule along with $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$

Q8. Find the tangent to the graph $y = \arctan 2x$ at the origin

Q9. A student suggests "The gradient of the graphs $y = \sin^{-1} x$ and $y = \cos^{-1} x$ are the same as the square root can be both positive or negative", comment on their statement

Advanced Skills

Q10. Find $\frac{d}{dx}(\cot^{-1}(x^2))$

Q11. Explain why $\sin^{-1} x = \operatorname{cosec}^{-1} \frac{1}{x}$ and find $\frac{d}{dx}(\operatorname{cosec}^{-1} x) =$

Q12. When performing the method we use $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$ why is this true

Q13. For $y = \arcsin\left(\frac{x}{3}\right)$

a) Show that $\frac{dy}{dx} = \frac{1}{\sqrt{9-x^2}}$

b) Hence evaluate the integral $\int_1^2 \frac{2}{\sqrt{18-2x^2}} dx$

Extension

Q14. By picking an appropriate trig substitution find $\int_0^1 \sqrt{9-x^2} dx$